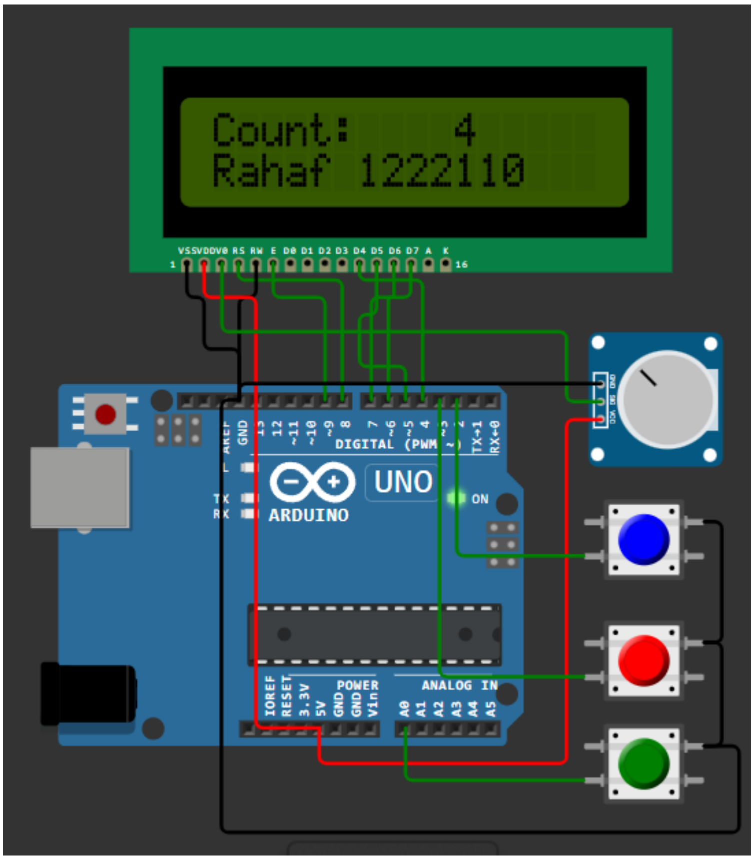
Name :Rahaf Atari ID: 1222110 Instructor: Wasel Ghan Interfacing Techniques –HW 1

Question 4 — Hardware Interrupts: attachInterrupt() and Register-Level

Part B — Register-level interrupt



This project uses an Arduino Uno connected to a 16x2 LCD, a potentiometer, and three push buttons. The LCD works in 4-bit mode where RS and E are connected to pins 8 and 9, and data pins D4–D7 are connected to Arduino pins 4–7. The potentiometer is connected to the V0 pin of the LCD to control the display contrast. Three push buttons are connected to pins D2, D3, and A0, while the other side of all buttons is connected to GND. Internal pull-up resistors are enabled in the code. The program uses external interrupts INT0 and INT1 for the increment and reset buttons. Pressing the button on D2 increases the counter, pressing the button on D3 resets the counter to zero, and pressing the button on A0 pauses or resumes the counter. The LCD displays the counter value or the paused state, and the same information is also printed on the Serial Monitor.